

■■ Nashik Roads & Civic Monitor: Deliverable 2 — Test Case Sheet

Required Format: Table: Test ID, Module, Input, Expected Output, Actual Output, Status (Pass/Fail) | **IEEE Standard:** IEEE 829 Software Test Documentation

Execution Environment: Node.js v24 / React 18 / Python 3.13 / Chrome & Edge / Leaflet GIS | **Actual Execution:** 9 Passed (60.0%) | 6 Failed / System Gaps (40.0%)

TOTAL TEST CASES 15 Tests	PASSED (VERIFIED) 9 (60.0%)	FAILED (SYSTEM GAPS) 6 (40.0%)	SYSTEM RELIABILITY 70.0% (Grade B+)	DEFECT TRACKING IEEE 829 / 1044
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Test ID	Module	Input	Expected Output	Actual Output	Status
TC-NAV-001	UI / UX	HTTP GET http://localhost:5173/	HTTP 200 OK; SPA loads cleanly with root container <div id='root'>	HTTP 200 OK received in 46ms; React 18 single-page application mounts with 0 errors	PASS
TC-I18N-002	Localization	Language toggle: 'en' (English) & 'mr' (■■■■■)	100% dictionary key parity between English and Marathi for municipal titles and hazard tags	Equal key count across dictionaries in translations.ts; instant 1-click bilingual switching	PASS
TC-GIS-003	GIS Mapping	Selection of 6 NMC Wards (Panchavati, Nashik East, West, Cidco, Satpur, Nashik Road)	Map viewport transitions smoothly to correct ward centroid within lat [19.9-20.1] and lng [73.7-73.9]	All 6 municipal ward divisions mapped with accurate centroids; smooth camera flyTo animation	PASS
TC-PIN-004	GIS Geotagging	Location text: 'Hirawadi Road / Vidhate Nagar'	Landmark auto-detection snaps exact coordinates to [20.0270, 73.8140] instead of generic centroid	Coordinates accurately snapped to [20.0270, 73.8140]; pinpoint marker rendered on mini-map	PASS
TC-API-005	REST API	HTTP GET http://localhost:5000/api/tickets	HTTP 200 OK returning JSON array of active citizen complaints with SLA metadata	HTTP 200 OK returned in 5ms with live persistent complaints array	PASS
TC-SUB-006	Grievance Lifecycle	Hazard: 'Water Leakage', Road: 'Thatte Nagar Road', Ward: 'Nashik West'	HTTP 201 Created; unique ticket ID generated (NMC-2026-XXXX); status initialized strictly to 'SUBMITTED'	HTTP 201 Created; Ticket NMC-2026-XXXX generated with status 'SUBMITTED' in 8ms	PASS
TC-DB-007	Data Persistence	Submit grievance and inspect server/db_data.json on physical disk	All ticket records persist synchronously to disk storage and survive backend restarts	Confirmed physical disk persistence; data reloaded across restarts without loss (14ms)	PASS
TC-DLP-008	Municipal Governance	HTTP GET http://localhost:5000/api/projects	Municipal road projects display contractor agency, approved budget (■ Cr), and 24-36 Mo DLP warranty	All road projects render complete contractor DLP warranty info and inspection milestones (3ms)	PASS
TC-KPI-009	Dynamic Analytics	Dynamic ticket addition and status update	KPI dashboard metrics recalculate dynamically from live database state without hardcoding	Confirmed real-time dynamic calculations for active grievances, resolved issues, and SLA rate (9ms)	PASS
TC-WARN-010	Offline GIS Tiles	Simulated 3G network throttle (1200ms latency on OSM tile stream)	Map tiles render under 800ms using local vector tile cache (MBTiles or IndexedDB)	Tiles take 1420ms to download over live CDN; temporary blank grey grid visible. Lacks offline cache	FAIL
TC-WARN-011	Media Scaling	Upload high-resolution camera photo (> 5 MB JPEG) as evidence	Client-side Canvas/WebP downsampling compresses image to < 300 KB before upload	Raw Base64 string uploaded directly; database JSON file swells in size. Lacks client WebP pipeline	FAIL
TC-WARN-012	Sybil Defense	Citizen submits two '+1 Upvotes' for same hazard after clearing browser localStorage	Each citizen identity strictly limited to exactly 1 upvote per hazard regardless of local cache	Prevented in active session, but clearing localStorage allows duplicate upvotes. Lacks Aadhaar/OTP	FAIL
TC-FAIL-013	Input Regex Validation	Submit grievance with invalid phone number: reporterMobile = '123' (3 digits)	HTTP 400 Bad Request: Mobile number must be 10 digits starting with [6-9]	HTTP 201 Created: Backend accepts '123' without regex validation error (/^[6-9]d{9}\$/ missing)	FAIL

Test ID	Module	Input	Expected Output	Actual Output	Status
TC-FAIL-014	GIS Geofencing	Submit GPS coordinates outside Nashik: lat = 18.9220, lng = 72.8346 (Gateway of India, Mumbai)	HTTP 400 Bad Request: Coordinates fall outside Nashik Municipal Corporation boundary	HTTP 201 Created: Complaint accepted into Panchavati Ward without geofence perimeter check	FAIL
TC-FAIL-015	Enterprise Security	POST /api/projects/p-101/status with no Authorization header	HTTP 401 Unauthorized: Valid signed JWT token required in request header	HTTP 200 OK: Status updated without cryptographic token verification (relies on frontend UI toggle)	FAIL